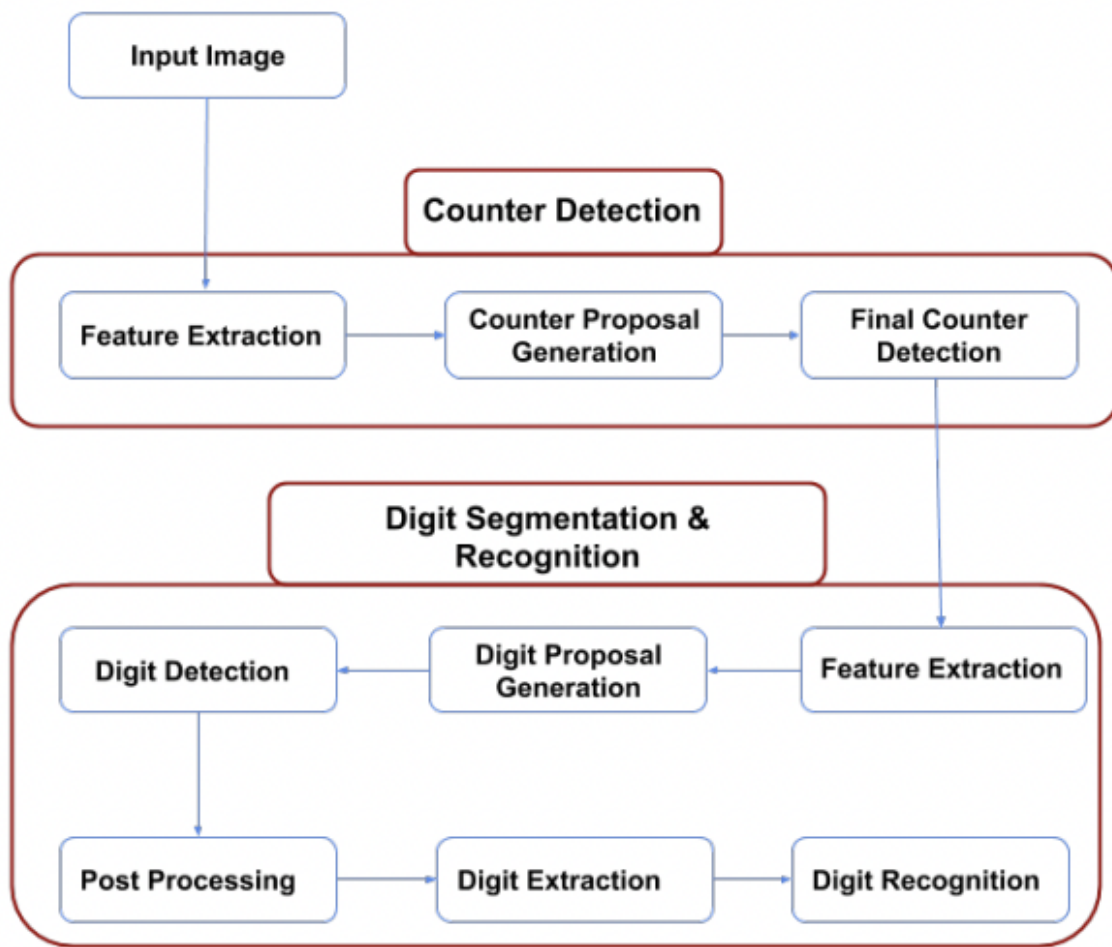
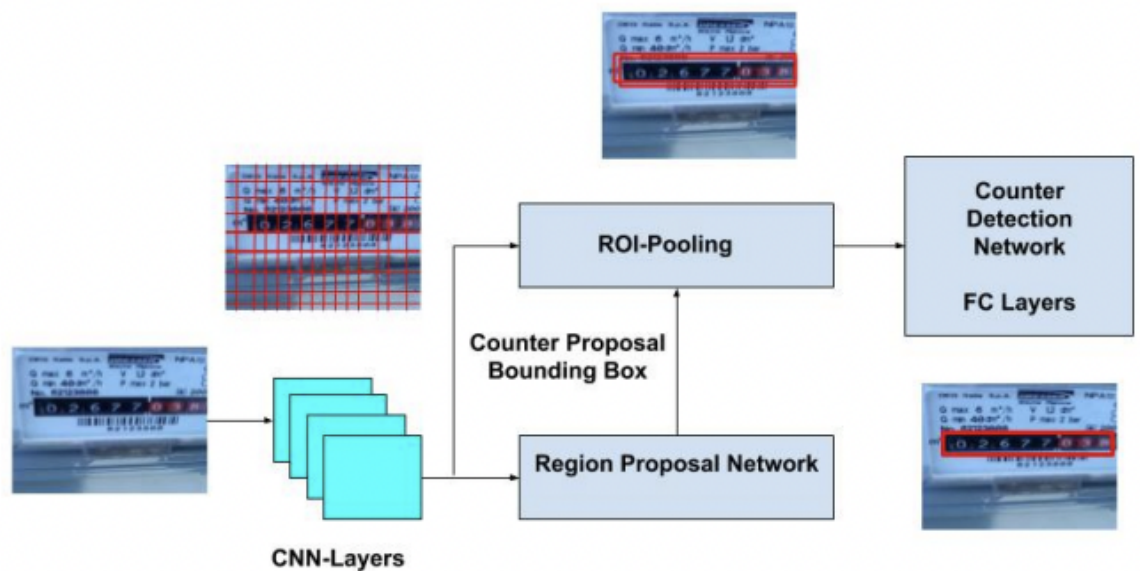




1. Mask-RCNN



a.

2. Counter Detection
 - a. Feature extraction
 - i. GoogleLeNet
 - b. Counter Proposal Generation
 - i. RPN, which takes features as input produces bounding boxes,
 - ii. $k = 16$ anchor boxes, and applied them at each position of the input feature map.
 - c. Final Counter Detection
 - i. utilizes the RoI-Align layer and set the pooling layer to 8×7 .
 - ii. To extract discriminative features, we use fully connected layers containing 2048 neurons with a dropout rate of 0.1.
3. Counter Segmentation and Recognition
 - a. Mask-RCNN is sorting digits and non-digits, as we want to detect each digit in a meter image. And predictions are made for digits and non-digits.
 - b. digits are removed, which are too small or too wide to be a digit. And regions are extracted, which may vary in y coordinate by amount s where s is the distance between y coordinates.
4. Loss Function and Training

$$L_{total} = (L_{cls} + L_{BBR})_{RPN} + (L_{cls} + L_{BBR})_{RCNN}$$

a.